AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target gene: eryCIV (UniProt **A4F7N3**) · *Saccharopolyspora erythraea* NRRL 2338 (NCBITaxon:405948) **Seed hypothesis under test:** eryCIV has **transaminase activity (GO:0008483)** — propagated by TreeGrafter/PANTHER, evidence type IEA, reference GO_REF:0000118 **Source selector:** `existing_annotations[2].function_hypothesis` in `genes/SACEN/eryCIV/eryCIV-ai-review.yaml`

Summary

The TreeGrafter/PANTHER graft assigned **transaminase activity (GO:0008483)** to eryCIV. My independent analysis finds this assignment **partially supported but too general**: eryCIV is a genuine pyridoxal-5′-phosphate (PLP)-dependent aminotransferase, but the propagated term stops at the broad *family* level when the protein in fact belongs to a well-defined *subfamily* carrying out a specific sugar C4-transamination reaction. Of the three characteristic phylogenetic-propagation failure modes the objective asks me to test, only **Failure Mode #1 (granularity / family-vs-subfamily)** applies. Failure Mode #2 (pseudo-enzyme / loss of catalysis) and Failure Mode #3 (within-superfamily mis-placement onto a functionally distinct neighbor) are both **affirmatively ruled out**.

The decisive evidence is active-site conservation. The DegT/DnrJ/EryC1 fold (Pfam PF01041) supports two divergent chemistries — C4 transamination versus C3,4-dehydration — within a single structural scaffold, making a wrong-subfamily graft a live risk that a tree graft cannot self-diagnose. I therefore directly tested the catalytic apparatus. EryCIV retains the catalytic PLP Schiff-base **lysine (Lys200)** in register with the catalytic lysines of the structurally characterized transaminases ArnB (Lys188) and WecE (Lys181), whereas the family members that *lost* transamination and became dehydratases (ColD, AscC) carry a **histidine** at that exact position. EryCIV has lysine, not histidine. It is **60.5% identical** to **DesI**, an X-ray-characterized dTDP-sugar C4 aminotransferase, far above any other characterized member. The AlphaFold model (mean pLDDT 96.5) independently places Lys200 and the PLP-anchoring Asp171 in a compact, well-ordered pocket with geometry consistent with a bridging PLP cofactor.

The practical outcome for curation is **retain-and-refine**: keep the transaminase assignment but make it more specific to a **dTDP-4-amino-4,6-dideoxy-D-glucose (C4) sugar aminotransferase** in desosamine biosynthesis — the direct ortholog of DesI. The legacy "3,4-dehydratase" database name that still clings to eryCIV is a pre-structural mis-label, contradicted by the active-site analysis, and should not be propagated.

Executive Judgment

Verdict: PARTIALLY SUPPORTED / TOO GENERAL.

- **Failure mode that applies: #1 (granularity / family-vs-subfamily).** The propagated term is correct in kind but under-granular.
- **Failure modes ruled out: #2 (pseudo-enzyme)** — all catalytic and cofactor-anchoring residues are intact and well-ordered; **#3 (within-superfamily mis-placement)** — EryCIV is on the transaminase side of the DegT/DnrJ/EryC1 split, not a ColD-type PLP dehydratase.
- **Single most decisive piece of evidence:** EryCIV retains the catalytic PLP Schiff-base **Lys200** (motif `SFHA-T-K-AVTS`), aligned residue-for-residue to the catalytic lysine of ArnB (Lys188) and WecE (Lys181). The non-transaminase member ColD carries **His** at this exact position — its hallmark loss-of-transamination substitution — and EryCIV does not. Combined with 60.5% identity to the structurally characterized C4 transaminase DesI, this fixes EryCIV as a genuine transaminase.

Independent Family / Function Assignment

Independent of the propagated PANTHER term, my own sequence and structural analysis assigns:

> **dTDP-4-keto-6-deoxy-D-glucose → dTDP-4-amino-4,6-dideoxy-D-glucose, a C4 sugar aminotransferase** in dTDP-D-desosamine biosynthesis.

- **Characterized homolog this is based on: DesI** (*Streptomyces venezuelae*, UniProt **Q9ZGH0**), the X-ray-characterized PLP-dependent C4 aminotransferase of the methymycin/

pikromycin desosamine pathway, solved in complex with its product dTDP-4-amino-4,6-dideoxyglucose (PMID: 17630700). EryCIV is **60.5% identical** to DesI over ~400 aligned columns — by far its nearest characterized neighbor. Erythromycin and methymycin/pikromycin share identical D-desosamine machinery, so eryCIV is the erythromycin-pathway ortholog of desI.

- **Domain architecture:** Pfam **PF01041 (DegT_DnrJ_EryC1)**, InterPro **IPR030967 (PLP_DesI)**; single-domain fold-type I PLP enzyme, 401 aa (SACE_0716).
- **GO term I would assign:** a **more specific** child of GO:0008483 on the **same branch** — functionally a "dTDP-4-amino-4,6-dideoxy-D-glucose transaminase" / sugar aminotransferase (EC 2.6.1.–). Granularity relative to the seed term: **more specific (same branch)** — not a sibling and not a different branch. Also well-supported: **GO:0030170 (pyridoxal phosphate binding)**, already present, and a desosamine/dTDP-sugar biosynthetic process term.

Key Findings

Finding 1 — The DegT/DnrJ/EryC1 transaminase active site is fully conserved; EryCIV is not a ColD-type dehydratase

The DegT/DnrJ/EryC1 family (PF01041) is a textbook case where one PLP-binding fold supports two divergent chemistries: **C4 transamination** (DesI, ArnB, WecE, EryCI/DesV type) and **C3,4-dehydration** (ColD, AscC). Because these diverged within a single fold, a graft could land eryCIV on the wrong side — the exact shape of Failure Mode #3 — and the tree alone cannot tell. I tested this directly with a Needleman–Wunsch global alignment (BLOSUM62) of EryCIV against characterized members, anchored on literature catalytic residues.

The two diagnostic PLP-handling residues are present and in register. The catalytic Schiff-base lysine, **EryCIV Lys200** (motif `SFHA-T-K-AVTS`), aligns exactly to **ArnB Lys188** (`SFHA-I-K-NITC`) and **WecE Lys181** (`SFHE-T-K-NYTA`) — the lysine that forms the internal aldimine with PLP and is absolutely required for transamination. Crucially, in the dehydratase members that lost transaminase activity this position is occupied instead by **histidine**: cross-alignment confirms the ColD residue aligned to ArnB Lys188 is His. **EryCIV has Lys, not His**, placing it unambiguously on the transaminase side of the split. The PLP pyridine-N1 anchor is likewise conserved: **EryCIV Asp170/171** in motif `FFDAAHA` aligns to **ArnB** `EDAAHAV`, the aspartate that hydrogen-bonds the protonated pyridine nitrogen of PLP.

Pairwise identities reinforce the placement by a roughly two-fold margin:

Characterized member	Function	% identity to EryCIV
DesI (Q9ZGH0)	C4 sugar aminotransferase	**60.5%**
ArnB (P77690)	sugar aminotransferase	37.5%
EryCI (P14290)	C3 sugar aminotransferase	36.1%
DesV (Q9ZGH4)	C3 sugar aminotransferase	32.5%
WecE (P27833)	sugar aminotransferase	32.4%
AscC (Q57174)	PLP 3,4-dehydratase	23.0%
ColD (G4WJD4)	PLP 3-dehydratase	**19.8%**

The ~2-fold identity gap between the nearest transaminase (DesI, 60.5%) and the dehydratases (≤23%), together with the Lys-vs-His discriminator, refutes Failure Mode #3.

Finding 2 — EryCIV's nearest characterized homolog is DesI, a C4 sugar aminotransferase: the seed term is correct but too general

EryCIV's closest characterized neighbor by a wide margin is **DesI** (60.5% identity over 400 columns). DesI is an X-ray-characterized PLP-dependent aminotransferase that converts **dTDP-4-keto-6-deoxyglucose → dTDP-4-amino-4,6-dideoxyglucose** (PMID: 17630700). Because erythromycin (in *S. erythraea*) and methymycin/pikromycin (in *S. venezuelae*) both deploy identical D-desosamine machinery, **eryCIV is the desosamine-pathway ortholog of desI**. This is precisely the granularity failure (Failure Mode #1): the propagated term names the broad *family* activity (any transaminase), while the protein carries out a specific C4 sugar-transamination step. UniProt annotation is concordant — EryCIV carries GO:0008483 (transaminase activity, IEA/TreeGrafter) and GO:0030170 (PLP binding), and the PLP-binding keyword is present.

This finding also surfaces the main **competing / historical** signal: EryCIV was originally deposited under a **"3,4-dehydratase"** name (it was said to "resemble the AscC 3,4-dehydratase"; PMID: 9353926). That label predates the structural era and is a **pre-structural homology carry-over** — the same vintage of error that for years left DesI itself mislabeled a

"4-dehydrase" despite being a proven aminotransferase. It is directly contradicted by the active-site analysis in Finding 1 (EryCIV has the transaminase Lys, not the dehydratase His) and must not be propagated by a curator.

Finding 3 — AlphaFold confirms an intact, well-ordered PLP aminotransferase pocket (not a degenerate active site)

To test Failure Mode #2 (fold retained but catalysis lost), I examined AlphaFold DB model **AF-A4F7N3-F1 (v6)**, overall mean **pLDDT 96.5**. The catalytic Schiff-base **Lys200** and the pyridine-N1-anchoring **Asp171** (motif `DAAHA`) co-localize in a single compact pocket, each modeled with very high local confidence (pLDDT 96–99). Measured geometry:

Residue pair	Distance	Interpretation
Lys200-NZ ↔ Asp171 carboxyl (OD1)	**6.98 Å**	Consistent with a PLP ring bridging the two anchors; Asp anchors pyridine N1
Lys200-NZ ↔ His174-NE2	3.15 Å	Pocket-lining His, H-bonding distance
Lys200-NZ ↔ Ser195-OG	4.52 Å	Pocket-lining Ser

Active-site pocket residues average **pLDDT 98.1**, indicating a **well-ordered, non-degenerate** catalytic site. There is no evidence of active-site erosion, loop disorder, or catalytic-residue substitution that would mark a pseudo-enzyme. The geometry is consistent with a PLP cofactor bridging Lys200 (internal aldimine) and Asp171 — i.e., an intact, catalytically competent aminotransferase active site. Failure Mode #2 is refuted.

Mechanistic Model / Interpretation

EryCIV operates in the **dTDP-D-desosamine** branch of the erythromycin deoxysugar machinery. Desosamine is one of two sugars (with L-mycarose/L-cladinose) appended to the erythronolide macrolactone; *eryC*-class mutants accumulate 3-α-mycarosyl erythronolide B, the signature of a block in desosamine attachment (PMID: 9563840). Within that pathway, EryCIV performs the **C4 transamination** step, installing the amino group that is later elaborated into the dimethylamino function of desosamine:

```
dTDP-4-keto-6-deoxy-D-glucose
           |
           |   EryCIV (= DesI ortholog)
           |   PLP-dependent C4 transaminase
           |   Lys200 forms internal aldimine with PLP;
           |   Asp171 anchors PLP pyridine N1;
           |   amino donor (L-Glu) -> PMP -> transfer to sugar C4
           v
dTDP-4-amino-4,6-dideoxy-D-glucose
           |
           |   (downstream N,N-dimethylation, etc.)
           v
       dTDP-D-desosamine  ---->  glycosyl transfer onto erythronolide
```

The family-level ambiguity that makes the graft under-specified — and its resolution:

```
             DegT/DnrJ/EryC1 fold (PF01041, fold-type I PLP)
                              |
       +----------------------+------------------------+
  TRANSAMINASES                                   DEHYDRATASES
(catalytic Lys at PLP site)                   (His at that position)
DesI, ArnB, WecE, EryCI, DesV                     ColD, AscC
       |                                              |
  EryCIV sits here  <-- Lys200 present            EryCIV is NOT here
  (60.5% id to DesI)                              (<=23% id; His discriminator absent)
```

The three findings converge on one coherent conclusion: EryCIV has the **right fold, the right subfamily, the right nearest characterized neighbor, and an intact catalytic apparatus**. The only defect in the propagated annotation is **resolution** — it stops at family-level "transaminase activity" rather than naming the specific sugar C4-aminotransferase reaction. This is a true-positive IEA requiring a **granularity refinement**, not a false positive.

Evidence Base

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	C li
1	PMID: 17630700 (Burgie & Holden 2007, *Molecular architecture of DesI*)	Structural (X-ray), direct	**Supports**	Is the nearest neighbor a transaminase?	DesI is a PLP-dependent aminotransferase; solved with product dTDP-4-amino-4,6-dideoxyglucose	*S. venezuelae* desosamine pathway	H th ch h E
2	This work (NW alignment, BLOSUM62)	Computational / evolutionary	**Supports**	Nearest characterized neighbor of EryCIV	60.5% identity to DesI; far above ColD (19.8%), AscC (23.0%)	UniProt sequences	H ga a M
3	This work (active-site mapping)	Computational / evolutionary	**Supports** (refutes Modes #2 & #3)	Catalytic Lys present (not dehydratase/ pseudo-enzyme)	EryCIV Lys200 + Asp170 conserved; ColD has His where transaminases have Lys	Aligned to ArnB K188, WecE K181, ColD	H ca ca w
4	UniProt A4F7N3 (database)	Database	**Qualifies**	Domain/ subfamily placement	Pfam PF01041 DegT_DnrJ_EryC1; InterPro IPR030967 **PLP_DesI**; PLP-binding keyword	*S. erythraea* SACE_0716	H In si ca
5	PMID: 9353926 (Summers et al. 1997)	Review/ database, mutant phenotype	**Competing / qualifies**	Historical "dehydratase" label	EryCIV "resembles AscC 3,4-dehydratase"; eryCIV mutant blocks desosamine	*S. erythraea* genetics	Pr st h ca co by Ly
6	PMID: 9563840	Mutant phenotype	**Supports (pathway role)**	EryCIV required for desosamine	eryC-class mutants accumulate 3-α-	*S. erythraea* in vivo	C pa ro

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	C li
	(Salah-Bey et al. 1998)				mycarosyl-erythronolide B (desosamine-deficient)		ex ch
7	AlphaFold DB AF-A4F7N3-F1 (v6)	Computational / structural	**Supports** (refutes Mode #2)	Active site intact, not degenerate	Lys200 + Asp171 form one well-ordered PLP pocket (pLDDT 96–99); Lys200-NZ↔Asp171 6.98 Å	Predicted structure	H co pr bo su

How the literature frames the conclusion. PMID: 17630700 provides the structural keystone: it establishes DesI — EryCIV's 60.5%-identical nearest neighbor — as a PLP-dependent aminotransferase and defines its exact reaction product, pinning EryCIV's specific function by orthology. PMID: 9563840 anchors the pathway role genetically, showing eryC-class disruptions produce the desosamine-deficient 3-α-mycarosyl erythronolide B phenotype. PMID: 9353926 documents the competing historical "3,4-dehydratase" label — the carry-over that motivated the active-site test and that this report recommends against propagating.

Active-Site / Placement Analysis (computed provenance)

Method: Needleman–Wunsch global alignment (BLOSUM62, gap −7) of EryCIV against characterized DegT/DnrJ/EryC1 members; catalytic positions anchored on literature residues (ArnB Lys188, WecE Lys181). AlphaFold geometry read from AF-A4F7N3-F1 v6.

Catalytic PLP Schiff-base lysine — INTACT:

```
                  catalytic residue v
EryCIV   ...S F H A T K A V T S...   (Lys200)
ArnB     ...S F H A I K N I T C...   (Lys188, catalytic)
WecE     ...S F H E T K N Y T A...   (Lys181, catalytic)
ColD     ...aligned position = His (NOT Lys)  -> dehydratase hallmark
```

Second PLP anchor (pyridine-N1 Asp) — INTACT: `EryCIV FFDAAHA (Asp170)` aligns to `ArnB EDAAHAV`.

Pairwise % identity to EryCIV (nearest-neighbor placement):

```
DesI  Q9ZGH0  60.5%   (characterized C4 transaminase)  <- nearest
ArnB  P77690  37.5%   (characterized transaminase)
EryCI P14290  36.1%   (characterized C3 transaminase)
DesV  Q9ZGH4  32.5%   (characterized C3 transaminase)
WecE  P27833  32.4%   (characterized transaminase)
AscC  Q57174  23.0%   (PLP dehydratase)
ColD  G4WJD4  19.8%   (PLP 3-dehydratase)            <- far
```

AlphaFold structural confirmation (AF-A4F7N3-F1, v6; mean pLDDT 96.5):

```
Lys200-NZ  <->  Asp171 carboxyl   6.98 Å   (spans a PLP ring; Asp anchors pyridine N1)
Lys200-NZ  <->  His174-NE2        3.15 Å
Lys200-NZ  <->  Ser195-OG         4.52 Å
pocket residues mean pLDDT = 98.1
```

Conclusion: Catalytic residues are **intact** and correctly spaced; EryCIV is unambiguously placed in the transaminase subfamily (DesI clade), not the dehydratase clade. The pocket is well-ordered, not degenerate — consistent with a catalytically competent aminotransferase.

GO Curation Implications

Recommended lead action (requires curator verification):

- **GO:0008483 transaminase activity (IEA): RETAIN, but MAKE MORE SPECIFIC.** Replace/augment with a child term reflecting a **dTDP-sugar C4 aminotransferase** (sugar aminotransferase; EC 2.6.1.–). The broad term is not wrong, merely under-granular for a protein with a 60.5%-identical, structurally characterized neighbor of known specific activity.
- **GO:0030170 pyridoxal phosphate binding: RETAIN.** The cofactor keyword and conserved PLP-binding residues confirm it.
- **Biological process:** flag any generic polysaccharide-biosynthesis propagation for the curator — the accurate BP is **dTDP-sugar / desosamine (deoxysugar) biosynthetic process**, not generic polysaccharide biosynthesis.

- **Do NOT** propagate the legacy "3,4-dehydratase" name — a pre-structural carry-over contradicted by the active-site analysis.
- This is a true-positive IEA with a **granularity refinement**. "Protein binding" is explicitly not an appropriate fallback.

Limitations and Knowledge Gaps

1. **No direct enzymatic assay for EryCIV itself.** The specific-function call rests on 60.5% orthology to DesI plus active-site conservation and genetics, not on a purified-enzyme kinetic assay of A4F7N3. Strong but inferential.
2. **AlphaFold is a prediction.** The 6.98 Å Lys200–Asp171 geometry and pocket ordering come from a predicted model (very high pLDDT, but not experimental coordinates); no PLP ligand is present in the model.
3. **Regiochemistry inference.** The exact C4-vs-C3 regiochemistry is inferred from DesI orthology; EryCI/DesV already account for C3 amination in related pathways, making C4 the parsimonious assignment for EryCIV, but this is not experimentally confirmed for A4F7N3.
4. **Reference-set and method scope.** Analysis is sequence/structure-homology based (programmatic UniProt + custom Needleman–Wunsch, linear gap, no curated MSA). The characterized-member panel is representative but not exhaustive; reference catalytic-residue numbering is from the literature while EryCIV numbering is from alignment.
5. **Substrate promiscuity.** PMID: 9563840 notes some ery deoxysugar enzymes "can accommodate alternative substrates," so the precise in vivo substrate channel is not fully pinned by sequence alone.

Proposed Follow-up Experiments / Actions

1. **Curator action (immediate):** Refine the EryCIV annotation from family-level GO:0008483 to the specific sugar C4-aminotransferase child term, citing orthology-to-DesI (PMID: 17630700) as ISS/ISO support; flag the "3,4-dehydratase" name as deprecated carry-over.
2. **Direct biochemical confirmation:** Express and purify EryCIV, reconstitute with PLP, and assay transamination of **dTDP-4-keto-6-deoxy-D-glucose** + **L-glutamate**, detecting dTDP-4-

amino-4,6-dideoxyglucose by LC-MS; include the **K200A** mutant as a loss-of-function control. This directly closes the one inferential gap.

3. **Cross-complementation genetics:** Test whether eryCIV complements a *desI* mutant of *S. venezuelae* (and vice versa), confirming functional orthology in vivo.
4. **Structural validation:** Solve EryCIV with PLP (or PMP + sugar-nucleotide), or 3-D superpose the AlphaFold model onto the DesI structure, to confirm the predicted Lys200/Asp171 geometry and substrate pose.
5. **Genome paralog check:** Confirm no second DegT/DnrJ/EryC1 paralog in *S. erythraea* has been confused with eryCIV in the annotation pipeline, ruling out orthology-assignment error at the genome level.

Bottom line

EryCIV (A4F7N3) is a **genuine PLP-dependent sugar aminotransferase** — the desosamine-pathway ortholog of DesI. The TreeGrafter/PANTHER graft identified the correct *kind* of activity (transaminase) but stopped at the family level. The decidable, evidence-backed action is to **retain the transaminase assignment and make it more specific** (dTDP-4-amino-4,6-dideoxy-D-glucose C4 aminotransferase). The pseudo-enzyme and wrong-subfamily failure modes are affirmatively ruled out by the intact catalytic lysine, the Lys-vs-His discriminator, and the well-ordered AlphaFold active site.
